

**AMENDMENT NO. 1 OCTOBER 2018  
TO  
IS 2386 (PART 3) : 1963 METHODS OF TEST FOR  
AGGREGATES FOR CONCRETE**

**PART 3 SPECIFIC GRAVITY, DENSITY, VOIDS,  
ABSORPTION AND BULKING**

(Page 6, clause 2.2.4) — Substitute the following for the existing clause:

**“2.2.4 Calculations**

Specific gravity, specific gravity based on saturated surface dry weight (*see* Note), apparent specific gravity and water absorption shall be calculated as follows:

$$\text{Specific gravity based on dry aggregate} = \frac{C}{B - A}$$

$$\text{Specific gravity based on saturated surface dry weight} = \frac{B}{B - A}$$

$$\text{Apparent specific gravity} = \frac{C}{C - A}$$

$$\text{Water absorption (percentage of dry weight)} = \frac{100 (B - C)}{C}$$

where

$A$  = weight of the saturated aggregate in water ( $A_1 - A_2$ ), in g;

$B$  = weight of the saturated surface dry aggregate in air, in g; and

$C$  = weight of oven dried aggregate in air, in g.

NOTE — This value of specific gravity based on SSD condition may be adopted for calculations for mix design as per IS 10262 : 2009 ‘Concrete mix proportioning — Guidelines (*first revision*)’.”

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(Page 8, clause 2.3.4) — Substitute the following for the existing clause:

#### “2.3.4 Calculations

Specific gravity, specific gravity based on saturated surface dry weight (*see* Note), apparent specific gravity and water absorption shall be calculated as follows:

$$\text{Specific gravity based on dry aggregate} = \frac{D}{C - (A - B)}$$

$$\text{Specific gravity based on saturated surface dry weight} = \frac{C}{C - (A - B)}$$

$$\text{Apparent specific gravity} = \frac{D}{D - (A - B)}$$

$$\text{Water absorption (percentage of dry weight)} = \frac{100 (C - D)}{D}$$

where

$A$  = weight of the vessel containing sample and filled with distilled water, in g;

$B$  = weight of the vessel filled with distilled water only, in g;

$C$  = weight of the saturated surface-dry sample, in g, and

$D$  = weight of oven-dry sample, in g.

NOTE — This value of specific gravity based on SSD condition may be adopted for calculations for mix design as per IS 10262 : 2009 ‘Concrete mix proportioning — Guidelines (*first revision*)’.”

(Page 11, clause 2.4.3) — Substitute the following for the existing clause:

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### “2.4.3 Calculations

Specific gravity, specific gravity based on saturated surface dry weight (*see* Note), apparent specific gravity and water absorption shall be calculated as follows:

$$\text{Specific gravity based on dry aggregate} = \frac{D}{A - (B - C)}$$

$$\text{Specific gravity based on saturated surface dry weight} = \frac{A}{A - (B - C)}$$

$$\text{Apparent specific gravity} = \frac{D}{D - (B - C)}$$

$$\text{Water absorption (percentage of dry weight)} = \frac{100 (A - D)}{D}$$

where

$A$  = weight of the saturated surface-dry sample, in g;

$B$  = weight of the pycnometer or gas jar containing sample and filled with distilled water, in g;

$C$  = weight of the pycnometer or gas jar filled with distilled water only; and

$D$  = weight of oven-dry sample, in g.

NOTE — This value of specific gravity based on SSD condition may be adopted for calculations for mix design as per IS 10262 : 2009 ‘Concrete mix proportioning — Guidelines (*first revision*)’.”